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Mythic dyes or mythic colour? New insight into the use of purple dyes on codices

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Abstract

The traditional knowledge concerning the use of Tyrian purple in Late Antique and Medieval purple codices - those precious biblical texts written with noble metal inks on parchment dyed or painted with purple colourants - is here updated in view of new analytical evidence. Recent literature reports the analysis carried out on some purple codices, suggesting that Tyrian purple has not, if ever, been used in their making. A large number of purple codices has been considered in this work to elucidate the nature of the purple colour. Results have been discussed within the frame of previous information, thus covering a vast majority of the purple codices presently identified. In most of the instances the use of less expensive dyes such as folium or orchil is suggested. Moreover, analytical results from a non-invasive spectroscopic approach have been definitely confirmed by micro-invasive surface-enhanced Raman analysis performed on micro samples of purple parchment taken from two 6th century codices.

Keywords (3-7)

purple codex; SERS; orchil; folium; Tyrian purple; Late antiquity

1) Introduction

The production of *purple codices*, the precious biblical texts written with noble metal inks on parchment dyed or painted with purple colourants, is cited in ancient treatises from at least the Hellenistic age.[1] Since purple codices were intended as highly prised books, produced under the commission of kings and emperors, their manufacture has long been debated among art historians with a focus on the *symbolic* rather than the *material* point of view. This resulted in scant interest in the materials actually employed for their manufacture. In addition, the high value and the fragility of these manuscripts have meant that very rarely the instrumental investigation of the colouring materials has been permitted. Given the lack of information, it has been commonly taken for granted that the purple hue was obtained by dyeing or painting the parchment with Tyrian purple, the highly prised dye extracted from molluscs, despite the absence of any analytical evidence.[2-5]

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Because of its extraordinary cost during the Roman period, associated with the symbolic significance of purple, the use of Tyrian purple was reserved for highly selective items such as luxury textiles worn by the nobility.[6] According to the *Editto di Diocleziano*, the price for one pound of purple-dyed silk was set at 150,000 *Denarii Communes*, resulting in the most expensive item among those traded in the Romani Empire; additionally, it was stated that purple silk was to be used only at the direction of the Emperor under penalty of death.[7]
From these premises, it seemed reasonable to assume the application of Tyrian purple for the colouring of the precious purple codices.
Tyrian purple has been the subject of many scientific studies[8] since the elucidation of its origin and chemical properties in 1833 by Bartolomeo Bizio[9] and of its structure in 1909 by structure in 1909 by Friedländer.[10] It has been identified in several instances on textile artworks,[11,12] less frequently in paintings[11,13] and never directly on decorated manuscripts.
The first analytical study on purple codices was performed in 1980 on the 6th century *Sinope Gospels* (Paris, Bibliothèque nationale de France, ms. Suppl. grec 1284).[14] Authors tentatively identified with GC-MS the presence of folium, the dye extracted from the *Chrozophora tinctoria* (L.) A. Juss. plant. Later, Wallert analysed a small fragment from a 5th century purple manuscript held at the Biblioteca Capitolare in Verona (Italy) with UV-visible absorption spectrophotometry and 3D spectrofluorimetry and identified alkanet, the dye extracted from the roots of the *Alkanna tinctoria* L. Tausch plant.[15] More recently, the application of non-invasive techniques resulted in information concerning colourants used in purple codices and the way they were applied.[16-25] None of these studies yielded direct evidence of the use of Tyrian purple; folium or lichen dyes (orchil) were identified instead.
It must be considered that no recipe exists in ancient treatises explicitly describing the use of Tyrian purple in the colouring of parchment, although the dyeing of textiles and textile fibres was described. The most important medieval treatises do not mention shellfish purple, i.e. a purple dye extracted from molluscs, in connection with parchment, while references to the application of substitutes of lesser value do exist, for example in the *Leyden Papyrus*,[26] the *Stockholm Papyrus*,[27] the *Manoscritto di Lucca*[28] and the 11th century manuscript 54 of the Biblioteca Capitolare in Ivrea (Italy), which contains one of the very few recipes specifically about the purple colouring of parchment. Recently Brun described a short 8th century treatise named *Conchylium*[29] that has recipes for the manufacture of purple codices; one of these recipes, *De tinctione porphyri* (On purple dyeing) describes dyeing with a colour obtained from a mollusc (conchylium), although it does not mention the material to be dyed.
Additional uncertainty arose from the ambiguity in these texts between the term ‘‘purple’’ and the actual compounds used to dye/paint with purple. Moreover, we shall consider that ancient treatises were frequently compilations obtained by copying more ancient documents, therefore unintentional mistakes, or intentional revisions, were possible while copying.
Therefore, although the traditional opinion concerning Tyrian purple is still popular among art historians, it should be probably reconsidered in view of new research and analytical evidences.
The identification of the specific colourants used for purple parchments is a relevant task, not only for the interpretation of the symbolic value of the colour in relation to the colouring matter, but also because information on the dyes is crucial for conservation, exhibition and storage of the very precious purple codices. We shall in fact consider that, whilst Tyrian purple is normally lightfast,[30,31] the same does not hold true for dyes obtained from lichens (orchil or cudbear) or for folium, that are definitely more fugitive.[32] For all these reasons, there is a need to widen the research in order to cover as many cases as possible.
A list of the still surviving purple codices produced from the 5th to the 11th century has been recently published by Quandt.[33] The list consists of sixty-nine records and includes all the presently identified manuscripts with folios that are dyed or painted purple over the entire

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surface. Efforts have been devoted in this work to elucidate the nature of the purple dyes in the vast majority of these codices, in order to offer new tools for the historical interpretation and for the conservation of these elaborated and costly books.
Table 1 collects all the purple codices that have been previously investigated and summarises the results of the diagnostic studies carried out so far.

Manuscript	Library	Century	Dyes
Codex Veronensis	Verona, Biblioteca Capitolare, cod. VI	5th	alkanet[15]
Codex Palatinus	London, British Library, Add. ms. 40107	5-6th	[a][34]
Codex Brixianus	Brescia, Biblioteca Queriniana	6th	orchil[23]
Codex Purpureus Petropolitanus	Rome, Biblioteca Apostolica Vaticana Vat. gr. 2305	6th	[a][34]
Codex Purpureus Petropolitanus	London, British Library, Cotton MS Titus C XV	6th	[a][34]
Codex Rossanensis	Rossano Calabro, Biblioteca Arcivescovile	6th	orchil[24]
Codex Sinopensis	Paris, BnF[b], Suppl. grec 1284	6th	folium[14]
Vienna Genesis	Vienna, Österreichische Nationalbibliothek, cod. theol. gr. 31	6th	orchil[20]
Zürcher Purpurpsalters	Zürich, Zentralbibliothek, RP 1	6th	orchil[18]
Coronation Gospels	Vienna, Kunsthistorisches Museum	7-8th	orchil, orchil/indigo[25]
Evangiles de Saint Riquier	Abbeville, Bibliothèque municipale, ms. 4	8th	folium[21,35]
Évangéliaire de Godescalc	Paris, BnF[b], NAL 1203	8th	folium or orchil[17,35]
Evangiles dits de Saint-Denis	Paris, BnF[b], Latin 9387	8th	folium or orchil[35]
Bible dite de Théodulfe	Paris, BnF[b], Latin 9380	9th	orchil[16,22,35]
Royal Bible	London, British Library, Royal MS I E vi	9th	[a][36]
Évangiles de Saint-Médard de Soissons	Paris, BnF[b], Latin 8850	9th	folium or orchil[35]
Codex aureus Epternacensis	Nürnberg, Germanisches Nationalmuseum, Hs. 156142	11th	orchil[19]

[a] Unidentified colourant. [b] BnF: Bibliothèque nationale de France.

Table 1. List of the purple codices analysed to date with the indication of the reported colourants.

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In order to extent the present knowledge on colouring materials and techniques that were employed to impart the colour, twenty-two purple codices have been considered in this new study with the aim of identifying the colourants and the way they were used. Non-invasive *in situ* techniques have primarily been used: X-ray Fluorescence Spectrometry (XRF), UV-visible diffuse reflectance spectrophotometry with optical fibres (FORS), Spectrofluorimetry and Optical Microscopy). However, in the cases of the *Vienna Genesis*[20] and the *Codex Brixianus*,[23] both 6th century manuscripts, further insight has been provided by Surface Enhanced Raman Spectroscopy (SERS) analysis performed on micro samples detached from the parchment.

2) Materials and Methods

FORS, XRF and spectrofluorimetric analysis were performed using the setup and instrumental settings described by Aceto et al.[23].

SERS analysis was performed by means of Ag colloidal pastes, according to the procedure described by Idone et al.[37] that modifies the method of Brosseau et al.[38] Nitric acid, hydrochloric acid, methanol, formic acid, silver nitrate and sodium citrate dihydrate were purchased from Carlo Erba reagents (Arese, Italy) while Ultra high quality (UHQ) water was obtained by a Millipore (Darmstadt, Germany) Direct-q 3 system. Raman measurements were performed with a Renishaw (Stonehouse, Great Britain) inVia micro-Raman spectrometer equipped with a 633 nm laser, 1800 lines/mm grating and a 100x Leica (Wetzlar, Germany) microscope objective to focus the laser beam onto the sample. Power at the samples was kept very low (< 300 µW) by a series of neutral density filters. Analysis of parchment samples was performed (1) directly by dropping 0.5 µL of silver colloidal paste on the parchment and (2) upon extraction of the dye by adding 50 µl of concentrated formic acid to a 1 mm^2 fragment of parchment kept at 40°C for 3 hours, then mixing 2 µl of extract with 2 µl of Ag colloidal paste.

A Leica (Leica Microsystems Srl, Milan, Italy) metallographic optical microscope DM2500 was employed to take high magnification (5x-100x) digital images of the parchment. The microscope is equipped with polarised light and a Leica DFC295 digital camera.

Paints and dye-baths for the purple colourants were prepared and applied as described elsewhere.[23]

3) Results

The list of the purple codices analysed in this current work is reported in Table 2. The manuscripts are dated from the 5th to 16th century and come from wide-ranging cultural areas (Near Eastern Asia, Byzantium, France, Italy, and Germany). It is therefore reasonable to assume that they could have been coloured in different ways or with different colourants.

Manuscript	Library	Century	Dyes
Codex Sarzanensis	Tortona, Biblioteca Diocesana	5-6th	orchil
Codex Brixianus	Brescia, Biblioteca Queriniana	6th	orchil, orchil/folium, orchil/indigo
Codex Argenteus	Uppsala, Universitetsbiblioteket	6th	orchil, orchil/indigo
Psautier de S. Germain	Paris, BnF[b], Latin 11947	6th	orchil
Vienna Genesis	Vienna, Österreichische	6th	orchil

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	Nationalbibliothek, cod. theol. gr. 31		
Codex Purpureus Petropolitanus	Vatican City, Biblioteca Apostolica Vaticana Vat. gr. 2305	6th	orchil
Codex Purpureus Petropolitanus	Vienna, Österreichische Nationalbibliothek	6th	orchil
Codex Sinopensis	Paris, BnF[b], Suppl. grec 1284	6th	orchil
Codex Aureus	Stockholm, Kungliga biblioteket, ms. A-135	8th	orchil
Evangiles dits de Saint-Denis	Paris, BnF[b], Latin 9387	8th	orchil, orchil/indigo, folium/indigo, brazilwood
Evangiles de Saint-Germain-des-Prés	Paris, BnF[b], Latin 11955	8th	orchil
Lectionnaire de Vérone	Paris, BnF[b], Latin 9451	8th	orchil
Salterio di Angilberga	Piacenza, Biblioteca Passerini-Landi	9th	orchil
Evangiles dits de Metz	Paris, BnF[b], Latin 9383	9th	orchil, orchil/indigo
Evangeliarium	Paris, BnF[b], Latin 1126	9-10th	orchil
Rotolo Borgiano	Vatican City, Biblioteca Apostolica Vaticana, Borg. gr. 7	11-12th	orchil, indigo, scale insect dye
Quintus Horatius Flaccus	Cambridge, King's College, ms. 34	15th	scale insect dye[a]
Libro d'Ore del Cardinale Carafa	Vatican City, Biblioteca Apostolica Vaticana, Vat. Lat. 9490	15th	orchil
Publii Ovidii Nasonis quae hoc in libello contintentur	Manchester, The University of Manchester Library, Aldine 3366.3	16th	folium[a]
Commentariorum de bello Gallico libri VIII	Manchester, The University of Manchester Library, Aldine 3553	16th	folium[a]
Prüm Lectionary	Manchester, The University of Manchester Library, Latin 7	16th	red ochre[a]
fragment of purple parchment	Vienna, Österreichische Nationalbibliothek, C2804	?	orchil

[a] A single leaf. [b] BnF: Bibliothèque nationale de France.

Table 2. List of the purple codices analysed (or reconsidered) in this work.

According to the appearance of the colour and to historical considerations - and considering the results obtained by previous analyses on other purple codices - the following colourants

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were considered: *Tyrian purple* (from several *Murex* snail species), *alkanet* (from the *Alkanna tinctoria*), anthraquinonic dyes (from *Rubia tinctorum* and other *Rubiaceae* species or from scale insect species), *folium* (from the *Chrozophora tinctoria*) and *orchil* (from several lichen species). The chemical structures of the main molecules present in these dyes are shown in Figure 1. Historical reconstructions of these colourants according to ancient recipes have been undertaken in the laboratories of the Università del Piemonte Orientale and the Università di Torino, creating dyed and painted reference parchment samples. Other parchment samples were dyed with *Lasallia pustulata*, *Roccella tinctoria*, *Hexaplex trunculus* and *Chrozophora tinctoria* at the Austrian National Library. The colourants were applied by immersion, by brush and by cotton swabs.

XRF analysis

The analytical investigation started with elemental analysis. Previous findings on paintings suggested that the identification of bromine in a purple coloured area could act as a marker for Tyrian purple.[39] Porter identified bromine in some early medieval codices by means of XRF analysis,[34] although the author commented that these results occurred in small details in which the use of a colourant as valuable as Tyrian purple was improbable. In most of the manuscripts analysed by XRF spectrometry in this study we found a small amount of bromine, according to K_α and K_β lines at 11.9 and 13.3 keV, but the intensity of the signal was too low when compared with a measurement performed on a reference sample of Tyrian purple paint (see Figure 2).

Indeed, as demonstrated in a recent study by Aceto et al.,[40] the identification of bromine in purple manuscripts cannot serve as a definitive evidence of the presence of Tyrian purple. Many coastal-sourced lichen species and *Chrozophora tinctoria* (L.) A. Juss, contain bromine in tiny – albeit detectable - amounts, possibly as a consequence of the exposure of these organisms to marine aerosol. The detection of bromine through elemental analysis, therefore, cannot be solely conclusive in distinguishing dyes from shellfish, folium or coastal-sourced lichen dyes. A database set of elemental measurements on lichens could possibly provide additional information, such as the possibility of distinguishing between coastal-based and terrestrial species, with the former having higher bromine content.

FORS analysis

Most of the FORS spectra collected from the parchment of the manuscripts considered in this work were very similar and highly reproducible. For these spectra, the two maxima in *apparent absorbance* coordinates[41] occurring at ca. 545 and 590 nm, allowed the identification of orchil or other lichen dyes (Figure 3).[42] The position of the less energetic maximum varied in the range 584-593 nm, possibly as a consequence of the variation in the lichen species from which orchil was prepared. The presence of folium cannot be totally excluded through FORS as its maxima are very close to those of lichen dyes; nevertheless the general shape of the absorption bands is different.

In some cases (*Codex Brixianus*, *Codex Argenteus*, *Coronation Gospels*, *Evangiles dits de Saint-Denis*, *Evangiles dits de Metz*), double colouring with orchil and indigo appeared to be present in sections of the books, based upon the additional band of indigo at ca. 655 nm, as can be appreciated in Figure 4. However, the exact sequence of application was unclear.

In very few cases other colourants or combinations of colourants were detected: folium/indigo and brasilwood in separate sections of the *Evangiles dits de Saint-Denis*; indigo and a scale insect dye in some parts of the *Rotolo Borgiano*; a scale insect dye in a leaf of the *Quintus Horatius Flaccus*; red ochre, the only case of an inorganic pigment, in a leaf of the *Prüm Lectionary*. All these colourants were identified through comparison with reference spectra taken on samples produced in laboratory.

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The presence of Tyrian purple could safely be excluded in all the manuscripts analysed, due to the absence of its typical absorption band at 526 nm.
In Table 3 the spectral features obtained from FORS analysis are summarised and compared with reference samples.

Spectrofluorimetry analysis
Spectrofluorimetry has already demonstrated to be valuable for detecting mixtures of folium and orchil (folium was indeed identified together with orchil in some sections of the *Codex Brixianus*[43]). Spectrofluorimetric analysis fully supported here the identification provided by FORS. In most spectra (Figure 5) a structured emission band was identified with a main maximum occurring at ca. 625 nm, characteristic of lichen dyes.[40,44] Where other colourants were suggested by FORS analysis, a coherent response was yielded by spectrofluorimetry.
In Table 3 the spectral features obtained from spectrofluorimetric analysis are summarised and compared with reference samples.

Colourant	Apparent absorbance maxima	Emission maxima
Tyrian purple	526, 640	no peak
Alkanet	525/560/600	625/660
Madder	510/540	600
Scale insect dye	525/560	625
Folium	545/580	595/610
Orchil	545/590	625
Indigo	655	no peak
mss. analysed	545-550/580-590/655[a]; [b]	623-627; [b]

[a] Additional band at 655 nm is only for the *Codex Brixianus*, the *Codex Argenteus*, the *Coronation Gospels*, the *Evangiles dits de Saint-Denis* and the *Evangiles dits de Metz*. [b] Different features in the *Quintus Horatius Flaccus* and the *Prüm Lectionary* and in separate sections of the *Evangiles dits de Saint-Denis* and the *Rotolo Borgiano*.

Table 3. Spectral features from FORS and spectrofluorimetric analyses.

SERS analysis
Micro-samples from the *Codex Brixianus* and the *Vienna Genesis* were used to obtain more in-depth information. A fingerprint identification of the dye was obtained by SERS analysis after extraction in formic acid. Using a colloidal dispersion of silver nanoparticles, well-structured SER spectra (Figure 6) were obtained. The SERS approach is particularly valuable for the detection of phenoxazones from lichens, as informative spectra cannot be obtained in conventional Raman mode with portable *in situ* or benchtop instruments due to the lack of response. The spectral features obtained here perfectly corresponded to the data reported in the literature for orchil.[40,45]
Fingerprint identification of orchil in the parchment of the *Codex Brixianus* and the *Vienna Genesis*, further support the identification of orchil in the other manuscripts analysed, considering that the spectral features obtained from FORS and spectrofluorimetric analysis were highly homogeneous within the group.

Dyeing vs. painting
Once the use of Tyrian purple on purple codices had been ruled out, a further question was: how were the substitutes (orchil, folium or other dyes) applied to the parchment? The scarce and incomplete information in the ancient literary sources is not surprising as most probably artisans wanted to keep their technical secrets hidden, considering the high symbolic value of purple codices.

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The recipe in the *Conchylium* manuscript describes a dyeing procedure,[29] but other treatises explicitly mentions the painting of the parchment. The recipe in the *Ivrea manuscript* describes painting parchment with orchil[28]. Similarly the *Ut auro scribatur*, a small treatise dating to the early Middle Ages, mentions parchment painted with orchil.[46] The same holds true for a recipe contained in *ms. Palatino 941* (15th century, Biblioteca Nazionale Centrale in Florence, Italy) which describes the use of *pezzuole* (small clothlets) soaked in the dye and rubbed on the parchment.[47]

The micro sample taken from the *Codex Brixianus* was partially used to prepare the cross section shown in Figure 7 at 500x magnification (top) together with a reference sample of parchment (ca. 100 µm thickness) dyed by soaking in orchil bath (middle) and painted with a solution of orchil in gum Arabic (bottom). Unfortunately, there is no visible difference between the two colouring techniques as in both cases the dye penetrated deeply into the parchment, making it impossible to suggest a procedure for the *Codex Brixianus*.

More informative were the images (Figure 8) taken at 90x magnification from the *Codex Aureus*, ms. Latin 9451 and ms. Latin 11947 and the *Rotolo borgiano*, and at 35x magnification from BnF ms. Latin 1126. In such cases, small losses of purple colour revealed the underlying uncoloured parchment, suggesting that the parchment of these manuscripts was painted, perhaps by rubbing the surface with a cloth soaked in the dye solution. It is interesting to note that for at least six hundred years (the manuscripts examined were made between the 6th and the 12th centuries) the same materials and same techniques continued to be used.

The experiments at the Austrian National Library showed that the colourants can be applied by immersion, by brush or by swabs. After dyeing the parchment has to be dried under tension. Orchil and folium can be used at room temperature.

4) Conclusions

This work significantly extends the number of purple codices that have been considered up to now for the scientific analysis of the dyes, and highlights the most common situation within a date range from the 5th to the 16th century. The results obtained indicated conclusively that the traditional opinion concerning Tyrian purple is mostly wrong and shall be strongly reconsidered. Despite the highly symbolic and material value of the purple manuscripts, the combined results of XRF, FORS and spectrofluorimetry analysis has in fact excluded the use of Tyrian purple in the colouring of the parchment, and lead us to affirm that orchil was instead used in most instances.

Besides this general picture, different materials have been detected in a few of the investigated codices, highlighting minor productions possibly imitating the appearance of the most valuable books.

The work has also fully demonstrated the merits of the non-invasive approach for the investigation of such precious testimonies of the past, as the unavailability of samples for micro-invasive procedures has prevented until now the chemical characterisation of the colourants.

The fascination of purple codices has leaded scholars to envisage the use of a mythic dye; non-invasive procedures have now highlighted a mythic colour instead.

5) Figure caption list

Figure 1. Chemical structures of the main molecules present in the dyes cited in this work.

Figure 2. Comparison of XRF spectra of the purple parchment of the *Coronation Gospels* (solid line) and of a reference Tyrian purple paint (dashed line).

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Figure 3. FORS spectra in *apparent absorbance* coordinates of the purple parchment of the *Codex Brixianus* and of reference orchil, folium, Tyrian purple, madder, scale insect dye and alkanet paints. Spectra are offset for clarity. Vertical lines highlight the two maxima at ca. 545 and 590 nm.

Figure 4. FORS spectra of a reference indigo paint and of indigo/orchil and indigo/folium mixtures.

Figure 5. Spectrofluorimetry spectra of the purple parchment of the *Codex Brixianus* and of reference orchil, folium, Tyrian purple, madder, cochineal and alkanet paints. Spectra are offset for clarity. The vertical line highlights the fluorescence maximum at ca. 625 nm.

Figure 6. SERS spectra of a micro sample from the *Vienna Genesis* (bottom line), a micro sample from the *Codex Brixianus* (middle line) and a reference orchil paint (top line). Spectra are offset for clarity.

Figure 7. Cross section of a micro sample from *the Codex Brixianus* (top), a parchment dyed with orchil (middle) and a parchment painted with orchil (bottom).

Figure 8. Photomicrographs from the *Codex Aureus* (90x, top left), BnF ms. Latin 1126 (35x, top right), ms. Latin 9451 (90x, bottom left) and ms. Latin 11947 (90x, bottom right).

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HIGHLIGHTS

- Purple codices are among the most precious book productions
- Tyrian purple has always reputed as the dye used for colouring in purple
- New analytical investigation revealed that lichen dyes (orchil) were used instead
- The use of orchil was continuous from late Antiquity until Renaissance

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